MECH MAINT- BOILER Department
SIMHADRI

Ref:SMTPP/MECH MAINT-/2022-23/LMI /226474 **Date:**27/06/2022(12:49:57)
Subject:REVISION OF LMI BTL FAILURE ANALYSIS AND CONTROL

REF.: SMPP/O&M/MM/PP/ Date: 27.06.2022

Subject: Review of LMI "BOILER TUBE FAILURE ANALYSIS AND CONTROL"

LMI No: LMI/OGN/OPS/MECH/049

The above approved LMI was revised in the month of Feb.'2022. The envisaged changes in LMI are incorporated as per details at revision index of LMI.

Put up for approval please.

Submitted by:

राहुल जैन - वरिष्ठ प्रबंधक

Rahul Jain - 013328 (SENIOR MANAGER ,MECH MAINT- BOILER) ,Simhadri
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On:27/06/2022(12:49:57) **Ref:**SMTPP/MECH MAINT-/2022-23/LMI /226474

May please accord approval for incorporating the changes as per revision in subject LMI.

Submitted by:

सुरेश बाबु कोल्ला - उप महाप्रबंधक

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On:30/06/2022(19:44:57) **Ref:**SMTPP/MECH MAINT-/2022-23/LMI /226474

The following changes (marked in bold) have been incorporated as per OGN in new LMI document attached as annexure (NEW LMI BTL doc)

1. Page no. 2 of 15 (Introduction)

2. Page no. 10 of 15 (Reporting of tube failure)

3. Page no. 15 of 15 (Point no. 34 added in C2 form)

Submitted for approval.

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

Om Prakash - 007105 (ADDL. GENERAL MANAGER ,MECH MAINT- BOILER) (OMPRAKASH08@NTPC.CO.IN, 9650994070) ,Simhadri
On:04/07/2022(09:05:29) **Ref:**SMTPP/MECH MAINT-/2022-23/LMI /226474

Submitted for approval of competent authority

Submitted by:

झिमलि धर - अपर महाप्रबंधक

Jhimli Dhar - 006460 (ADDL. GENERAL MANAGER ,EEMG)
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On:06/07/2022(10:13:18) **Ref:**SMTPP/MECH MAINT-/2022-23/LMI /226474

May please be accorded approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoor - 005637 (GENERAL MANAGER ,MAINTENANCE)
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On:17/07/2022(08:25:48) **Ref:**SMTPP/MECH MAINT-/2022-23/LMI /226474

कृपया अनुमोदन प्रदान किया जाए।

May be approved please.

Submitted by:

राजेश कुमार झा - महाप्रबंधक

Rajesh Kumar Jha - 003323 (GENERAL MANAGER ,O & M)
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On:17/07/2022(10:28:37) **Ref:**SMTPP/MECH MAINT-/2022-23/LMI /226474

Approved & Submitted by:

गिरीश चन्द्र चौकसे - महाप्रबंधक

Girish Chandra Choukse - 002907 (GENERAL MANAGER ,HEAD OF PROJECT)
(GCCHOUKSE@NTPC.CO.IN, 9425178006) ,Simhadri
On:18/07/2022(12:01:10) **Ref:**SMTPP/MECH MAINT-/2022-23/LMI /226474

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/049/

ISSUE NO.1; REVISION NO.1; DATE: MAY 2022

BOILER TUBE FAILURE ANALYSIS AND CONTROL

APPROVED FOR

Implementation by ....

HOP, SIMHADRI

Date ------------------------

1.0INDEX

2.0INTRODUCTION

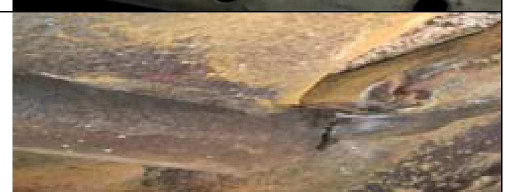
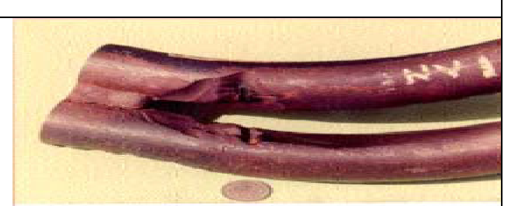
Boiler tube failures analysis is the first step towards control of boiler tube failures. NTPC in line with EPRI literature has instituted a mechanism of BTF analysis and its control. Repeat failures due to known mechanism are the first target of BTL reduction programme. They severely dent morale of the workforce and causes significant financial loss to the company. Increased size of the units has increased the loss multiple fold and therefore there is more management concern to control this preventable loss.

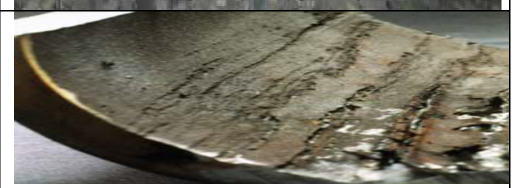
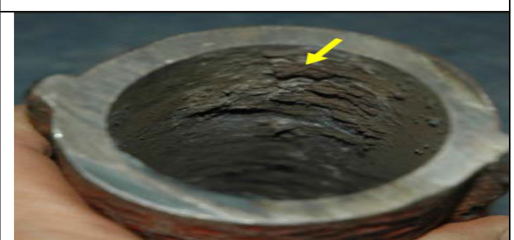
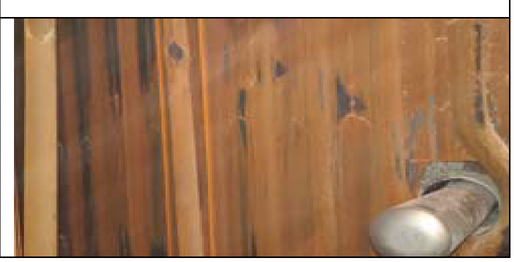
Boiler tube damage accumulation and eventually final failures are due to various reasons which are sometimes very complex and need in depth analysis and therefore time consuming. Boiler tube failures due to Build Quality Deviation from original with respect to approved design during erection & commissioning, operational issues viz: Metal temperature fluctuations & excursions, Chemical Parameter Deviations, Maintenance Induced viz: improper restoration of attachment & supports, misalignment, improper/incomplete fin welding, material mix up, improper preheat and PWHT during erection, tube starvation due to foreign material inside headers/tubes are regularly coming up even after availability of specific information to avoid them.

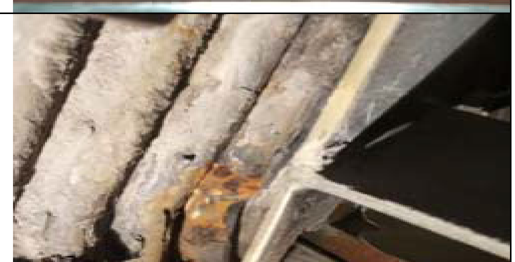
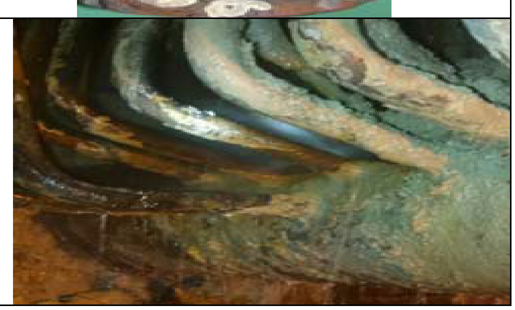
Tube failures because of short term overheating due to Start up Deviations and as well as “Mechanical Design Failures” of attachments due to thermal transients causing fatigue have become new area of challenge as cyclic loading requirement of the unit as per grid demand and **introduction of AGC & FGMO** has increased substantially. **It is desirable to keep MS temperature variation Standard Deviation less than 3.5.**

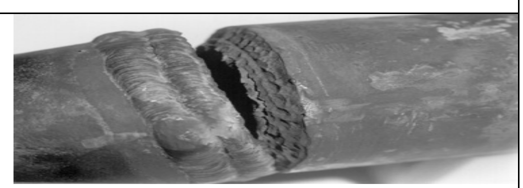
Chemical parameter deviations during startup are quite critical from point of view of supercritical boilers using lot of Stainless steel which is highly susceptible for stress corrosion cracking. Exfoliation of SS tube scales due to wide temperature fluctuations and excursions during start up and shut down are critical area of concern in supercritical boiler. This category has been increasing in recent years with the increase in starts-stops & ramp Up/down of the units in view of grid requirement. Frequent revisions in load schedule has further aggravated the fatigue stress on the machines. Boiler components are significantly affected by the thermal cycles and their severity. There is good evidence that inter-ligament cracking on Superheater outlet headers is exacerbated by such transients and the increase in the incidence of Type IV cracking. Attachment failures in water wall and superheaters connectors are showing increasing trend in view of high thermal transients during cyclic loading therefore comprehensive action plan from Design > Manufacturing > Storage > Erection> Commission > Preservation and subsequent Operation & Maintenance must to be followed to encompass all these steps. Extensive use of advance NDT during periodic maintenance has to be part of BTL control system.

Controlling BTL reduces secondary damages in the boiler due to steam washing and overheating downstream of the failure caused by the practice of “running on” with a leak to get an opportunity for a suitable outage period. This LMI provides the list of all guidelines issued by COS to control the boiler tube failures. BTL mechanisms for ready reference as per EPRI guidelines are as follows:

Water Touched Tubes	Leakage Appearance	Causes	Location	Sample photo
A. WATER WALL THCIK EDGE FRACTURE				
Corrosion Fatigue	Thick Edge/Pin Hole, Circular Crack/blow out	Poor Water Chemistry, High Do, Phosphate Hide Out, Improper Lay Up, Cycling,	High Stress Attachments, bends at neutral axis, generally at cold side of tube in Water wall & Economizer	
Hydrogen Damage	Thick Edge Window Blow Out	Condenser leak, Poor Water Chemistry, Start Up,	High Heat Flux Area On Hot Side of water wall, near flow disruptions	
Thermal Fatigue Supercritical Walls	Thick Edge Cracks From Out Side	Water Wall Deposits, Poor Water Chemistry, High DO, Improper wet to dry changeover	Maxi Heat Flux area of Water wall	
Thermal Fatigue (Economiser Inlet)	Thick Edge Crack /Pin Hole At Stub Toe	Frequent Start Up & Cycling,	Eco I/L Header Near Feed Water I/L Point	
Low Temperature Creep	Thick Edge Cracks At Outside Surface	Residual stress of Cold forming, Deformation in bending	Tube Bends With High Stress	
Chemical Cleaning Damage	Zigzag, rough undercut pits	Improper cleaning	Water Stagnation zone after cleaning and flushing,e.g. horizontal/ slant tubes	
Fatigue in water cooled circuits	Thick Edge	Header and feed water temperature difference	Near Attachments, Jammed Sliding Attachments	

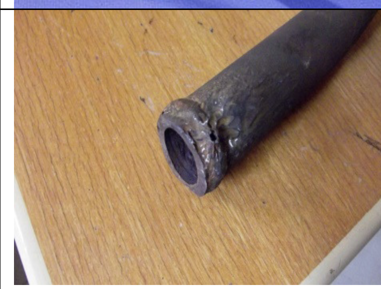
Water Touched Tubes	Leakage Appearance	Causes	Location	Sample Photo
B WATER WALL THIN EDGE FRACTURE				
Fly Ash Erosion	Thin Edge Fish Mouth	Coal Quartz, Unbalanced air flow and excess air flow, Misalignment	Near Side and Rear Walls, II Pass, Plugged Passages, Out Of Alignment tubes	
Acid Phosphate Corrosion	Thin Edge Leak/ Split	Phosphate Hide Out, Poor Water Chemistry , High Do, Black Water	High Heat Flux Area On Hot Side of water wall	
Caustic Gauging	Thin Edge Leak/Split	Poor Water Chemistry, High Do, Caustic>2ppm	High Heat Flux Area On Hot Side of water wall	
Fire side corrosion	Thin edge fish mouth	SADC Mismatch, Coal Composition	Reducing Zone/ High Heat Flux Area in burner zone	
Erosion-Corrosion Economizer Inlet **(FAC)**	Thin Edge/ Wall Thinning From Inside	Poor Water Chemistry, High DO	Eco I/L Header stub tubes, Near Feed Water I/L Point	
Soot Blower Erosion	Thin Edge Fish Mouth	Poor soot blower maintenance/ protection	Near Soot Blower, especially tubes perpendicular to soot blower or out of plane	

	Water Touched Tubes	Leakage Appearance	Causes	Location	Sample Photo
	Short Term Overheating	Generally Thin Edge/ Sometimes Thick Edge	Low drum level, foreign material in tube, orifice plug gage	Highest Heat Flux Location like Burner top zone and Goose neck area fire side tube.	
	Pitting In Water Cooled Tubes	Pitting On internal surface of Water Wall	Poor water chemistry/ preservation	At horizontal tubes and coils	
	Coal Particle Erosion	Thin Edge	Damaged Burners tips	Burner transition tubes/ throat and qual region tubes	
	Falling Slag Erosion/ Damage	Thin Edge	Improper air distribution, Improper S panel support	Slopping Wall Tubes	
	Acid Dew Point Corrosion	Thin Edge	Low flue gas exit temp (< 120 deg)	Last stages Of Economiser	

Water Touched Tubes	Leakage Appearance	Causes	Location	Sample Photo
C THICK EDGED FRACTURE				
Long Term Overheating Creep	Thick Edge, with adjacent multiple fine cracks	Partial blockage, Material inadequacy	Highest Temp Near Material Transition At Gas Facing/ Cavities of Superheater and reheater and waterwall in burner area or just above it.	
Low Temp Creep Cracking	Thick Edge	Residual stress of Cold forming, Deformation in bending or excess service stress	Low Temp Region Tube Bends Intrados/ Outside At High Stress Points	
Dissimilar Metal Weld	Thick Edge	Improper WPS/ consumable/ Creep	At Fusion Line Cracking On Low Alloy Side of Super heater and Reheater or penthouse area DMW	
Stress Corrosion Cracking	Thick Edge/Pin Hole	Condenser leak, thru spray water	High Stress Locations,Bends, Welds, Attachments mostly but not limited to Austenitic S.S	
Fatigue In Steam Touched Tubes	Thick Edge	Fatigue life exhaustion/ Residual stress	At Bends And Attachments, stub connections at Header Ends	

Water Touched Tubes	Leakage Appearance	Causes	Location	Sample Photo
Graphitisation In C /C-Mo Tubes	Thick Edge	Material inadequacy, Improper WPS	Adjacent To Weld Fusion Line HAZ, in low temp zone of SH & RH	
Fire Side Corrosion	Thin Edge	Improper SADC/ Damaged coal nozzle	Highest Temp Tubes, Leading Tubes, Cavities, Out of Alignment	
Short Term Overheating	Thin Edge	Condensate in tube bend, Excessive spray	Near Bends or at Material transition	
SH/RH Soot Blower Erosion	Thin Edge	Improper pressure/ nozzle size/ protection	First Tube Facing LRSB, tubes closest to LRSB	
Rubbing Tubes/ Fretting	Thin Edge	Improper support/ locking/ excessive spray	Metal to metal Contact in Spacer/ hanger tube to coil tube	

Water Touched Tubes	**Leakage Appearance**	**Causes**	**Location**	**Sample Photo**
Pitting corrosion	Pin hole	Poor preservation/ venting	Condensate At Bends And Sagging Portion In Horizontal Tubes	
SH/RH Chemical Cleaning	Pin hole	Incomplete draining/ flushing out after chemical cleaning	At the bottom bends of the pendant coils	
Fly Ash Erosion	Thin Edge	High negative furnace draft/ blocked passage in Flue gas ducts/	Back Pass Region Bends Near back pass C header, Convective zone side coils or Reheater tubes near the roof	
Maintenance Damage	Can be of any shape or size	Induced damage or ingress of foreign material during maintenance activities, e.g. gas cutting of vertical tubes	Previous maintenance location, mainly near weld joints or attachment joints	

Water Touched Tubes	Leakage Appearance	Causes	Location	Sample Photo
Material Flaws	Cracks, blow offs	Wrong material use, manufacturing flaw (Forging lap/ lamination)	Mostly at but not limited to bends	
Welding Flaws	Pin holes, cracks	It can be due to wrong procedure, wrong joint preparation or fitment, improper consumable selection or use	Tube joint	

3.0 SCOPE

The scope of this LMI is to detail the records to be maintained on the incidence of boiler tube failure and methods that are to be adopted for selection and handling of failure samples.

4.0 REFERENCE DOCUMENT

Operation Guidance Note COS-ISO-00-OD/OPS/MECH/049 ,Rev 1,May 2022 circulated by operation services-NTPC corporate center.

5.0 SUPERSEDEDDOCUMENT

PREVENTION & CONTROL OF BOILER TUBE FAILURES REV:0 DATE: Feb -22

6.0 SPECIFICINSTRUCTIONS

The following approach is to be followed to attain the objectives of this LMI

a) Reporting and recording of tube failure in the designated forms
b) Selecting and handling failure samples properly for effective failure analysis

7.0METHODOLOGY OF IMPLEMENTATION AND RESPONSIBILITY CENTERS

REPORTING TUBE FAILURE AND RECORDING ON-SITE EXAMINATION

A) TUBE FAILURE NUMBER

A tube failure no .shall be given to each tube failure at the station as per the following format.

BTF No/Station/Unit/Date of failure

This tube failure no. will be mentioned in Pradip File subject and in each document related with this tube failure and should be strictly adhered in all reports, documents and drawings.

Responsibility-HOS(PP)

B) REPORTING OF TUBE FAILURE

Different types of information is required for the initial analysis of a tube failure. These are to be recorded in the boxes provided on Format No. OS/BLR/TF/002, the NTPC Boiler Tube Failure Record (C-2Form) and shall be sent to corporate OS. **Initial on-site prior to repair or removal of the failed tube pictures to be shared in NTPC Boiler group with initial BTL information as a part of form C2.**

Responsibility-HOS(PP)

C) SELECTION AND HANDLING OF SAMPLES

i) Sample for failure analysis should include the ruptured portion. The sample should be long enough to include reference material to allow comparison which can distinguish between local phenomena such as localized overheating and a more general phenomenon such as long term overheating.

Responsibility-HOS (PP)

ii) Tube samples should be preferably removed using a saw with cuts at least 300 mm from the point of rupture. Torch should not be used for cutting as overheating can change the microstructure of the steel and destroy primary evidence.

Responsibility-HOS(PP)

iii) The samples are to be handled with care so that as much information as possible about failure mechanism is preserved. The scales on the exterior of tube sample should not be cleaned with wire brush.

Responsibility-HOS(PP)

iv) Samples should be marked in a manner that their position and orientation in the plant are defined. Direction of steam/ water flow should be marked. Any unusual event noted prior, during / just following the failure mechanism should be noted along with the sample. Before removing any samples, if the circumstances are unusual a photographic record of failed tube with location and position of soot blowers/burners should be maintained. The samples should be packed carefully so that they are not damaged during transportation.

Responsibility-HOS (PP)

8.0 EXCEPTIONREPORTING

The exceptions encountered during implementation of this system as compared to laid down procedure as above are to be reported as per the following format and to be taken up with corporate OS Mech.Mtc.(Pr.Parts group)

Description of exception	Unit No.	Location of exception	Remarks

9.0 RESPONSIBILITIES

The responsibility of carrying out different tasks in relation to prevention, analysis and control of tube failures as detailed in the LMI are with HOD(BMD)

10.RECORDKEEPING

The records to be maintained under this LMI include
a) Compliance record of the activities detailed under this LMI
b) C-2FORMS
c) Exception reporting

The above records are to be maintained by HOS-Pr. Parts Group.

11.REVIEW

This document shall be reviewed by Head of The Project on two-yearly basis.

REVISION INDEX

SL NO.	Rev. no.	Date	Details	Approved by
1	1	27.06.2022	1) Introduction changes 2) Introduction (failure investigation) 3) C2 form changed	

DISTRIBUTION LIST

SI. No.	Date of issue	Issued to	Copy no.
1		HOD (O&M)	
2		HOD (MAINT.)	
3		HOD (BMD)	

NTPC	COMPILANCE RECORD	SIMHADRI
TITLE: BOILER TUBE FAILURE ANALYSIS AND CONTROL REV: 1 DATE: MAY 22		
REF NO. LMI/OGN/OPS/MECH/049/		PAGE : 14 OF 15

UNIT NO.	ACTIVITY	COMPLIANCE
I/II/III/IV		

BOILER TUBE FAILURE REPORT (C-2 FORM) (TO BE SEND WITHIN 36 HR OF SYN)

C2 Submission date

BTL identification, Pradip file name & C2 File Name to be same as: BTF No of financial year/ Station/Unit/Date

1 Station:
2 Unit No.
3 BTL Date
4 Last Unit Overhaul
5 Last Unit Start Up
6 No of Start Up Since last OH
7 No of BTL Since Last OH
8 Total No BTL in Unit in last 10 yrs.
9 **Unit Syn**
10 Generation & Financial loss on &
11 Last Condenser tube leakage in Unit
12 FW ACC Maxi/AV in last 24 hrs.
13 DO Maxi/AV in Condensate in last 24 Hrs.
14 FW pH Min/ AV in last 24hrs
15 ASLD ALARM at
16 Unit Stopped at
17 No of times unit Acid cleaned
18 **Last Acid cleaning date**
19 BTF INSPECTED BY

BTF COMMITTEE HEAD		MTP		FQA	
CHEMISTRY		OPERATION		O&E	

BOILER TUBE FAILURE LOCATION

20 **FAILED TUBE LOCATION MARKED ON UNIT MATERIAL DIAGRAM (insert here) showing tube material**
(CLEARLY IDENTIFYING TUBE NO, COIL NO, TUBE DIMENSION, TUBE MATERIAL AND X, Y, Z COORDINATES)

21 **FAILED TUBE LOCATION MARKED ON BOILER GENERAL ARRANGEMENT DRAWING (insert here)**

DESCRIPTION OF BOILER FAILED TUBE

22 **PHOTOGRAPHS OF LOCATION OF FAILURE BEFORE START OF REPAIR WORK (insert here)**

23 **PHOTOGRAPHS OF FURNACE FOR CLINKERING /ASH DEPOSITION (insert here)**

24 **Soot blowing and LRSB Operation** **Before S/D** **SB** **LRSB** **After S/D** **SB** **LRSB**

25 **PHOTOGRAPHS OF PRIMARY FAILED TUBE (insert here)**

26 **PHOTOGRAPHS OF REPAIRED LOCATION AFTER RESTORATION WORK (insert here)**

27 **DETAILS OF REPAIR WORK (SPECIFY WELD JOINT DETAILS, COIL NO/ TUBE NO, RESTORATIONWORK DETAILS WPS, SR ETC.)**

28 **DETAILS OF INSPECTION OF ADJACENT AND IDENTICAL LOCATIONS**

29 **HISTORY OF BTL & Work done IN FAILED TUBE/COIL/LOCATION IN PAST FIVE YEARS**
(Attach Boiler Sketch Side & Front View with identifying BTL location/coil no/tube no/date)

30 **Failure Mechanism**

31 **BTF Committee Initial Observation W.R.T Root Cause of Failure & Action Plan**

Tube thickness during last overhaul: … mm 2. Adjacent tubes thickness during last overhaul ….mm
3.In case of SH/RH insitu oxide during last OH …. micron 4. Adjacent tubes insitu oxide during last overhaul …. micron Action Plan ….

32 **Action Plan BOQ number in SOW and Planned Implementation Date**

33	Locat ion	Boiler cooling	Water washing	Scaff folding	Inspect ion	Repair Time	Radiography	Rest orat ion	Scaffolding removal	Filling and checking	Dumping	Lit up clearance	Act ual Lit up	Syn at	Bar to Bar time
Actua															
Estimated (OS)															

34 **Job completion final inspection by Area Engineer (Name) & Record liability if any**

35 **STAITON COMPLIANCE SCHEDULE WRT TO COS GUIDELINES /RECOMMENDATIONS: SOW BOQ NUMBERS AND TIMELINE**

FORMAT NO. OS/BLR/TF/002 REV.10

(Digital Signature of BTF Committee Head) DATE
NAME:
DESIGNATION:
Mobile No

Station BTF Committee head to move C2 file in Pradip to COS BTF Coordinator in agreement with BTF committee
C2 copy in email to be provided to all concerned at station, region, COS KT member and Head COS Boiler
ALL BTF SAMPLES TO BE SEND TO COS BTF COORDINATOR WITHIN 07 DAYS